ASSIGNMENT SUBMISSION MODULE 1

For this project, I applied data mining techniques to a real-world Kenyan voting dataset containing 20,000 voter records across 13 attributes spanning the 2013, 2017, and 2022 election cycles. The project workflow followed an end-to-end data mining pipeline, from data acquisition to clustering and analysis.

1. Dataset Acquisition and Cleaning
   - The dataset was sourced from Kaggle: “Kenya Voting Patterns and Top Regions by Voting Bloc.”
   - Initial exploration revealed no missing values or duplicates, ensuring 100% data quality.
   - Categorical features (e.g., Region, Party) were encoded numerically for algorithm compatibility.
   - Outlier analysis confirmed that extreme values were within expected demographic ranges.
2. Exploratory Data Analysis (EDA)
   - Conducted univariate and bivariate analyses to understand distributions and correlations.
   - Identified patterns in voter turnout, regional party loyalty, and cross-party voting.
   - Feature correlations were visualized, showing meaningful relationships between age, gender, region, and party affiliation.
3. Dimensionality Reduction
   - Applied Principal Component Analysis (PCA) to reduce the feature space from 7 selected features to 6 principal components, retaining 93.15% of the variance.
   - PCA facilitated faster computation and improved interpretability for clustering.
4. Algorithm Selection: K-Means Clustering
      - Justification for K-Means:Dataset is numerical and medium-sized, suitable for distance-based clustering.
      - The objective was to identify natural voter archetypes and regional patterns.
      - K-Means is computationally efficient, scalable, and produces easily interpretable clusters.
      - Alternative algorithms considered:Hierarchical Clustering: Too slow ($O(n^2)$) for 20,000 records.
      - DBSCAN: Sensitive to parameter selection, less effective for varying-density clusters.
      - Gaussian Mixture Model (GMM): Assumes Gaussian distributions, not guaranteed in this dataset.
      - Spectral Clustering: High memory overhead and unnecessary for numerical features.

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5. Clustering Implementation and Evaluation
   - Determined the optimal number of clusters using the Elbow Method and Silhouette Analysis, resulting in 4 distinct clusters.
     - Cluster evaluation metrics:Silhouette Score: 0.1587 (acceptable for overlapping clusters)
     - Davies-Bouldin Index: 1.7764 (indicating reasonable separation)
     - Cluster profiles identified:UDA Male Voters - 25%, high turnout
     - Non-Voters - 23.6%, no turnout
     - UDA Female Voters - 25.2%, high turnout
     - ODM Regional Voters - 26.1%, ~96% turnout
6. Analytical Insights
   - Clear geographic and gender-based voting patterns were revealed.
   - Voter archetypes can inform targeted campaigns and resource allocation.
   - Cross-party voting behaviors were observed, highlighting potential swing regions.